

```
!pip install ultralytics
```

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Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (11.3.0)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (6.0.3)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.32.4)
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.16.3)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.11.0+cu128)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (0.26.0+cu128)
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (5.9.5)
Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.35.2)
Requirement already satisfied: nvidia-ml-py>=12.0.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (13.610.43)
Collecting ultralytics-thop>=2.0.18 (from ultralytics)
```

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  Downloading ultralytics_thop-2.0.20-py3-none-any.whl.metadata (14 kB)
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Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.3.0)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (4.53.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.4.7)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (24.1)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (3.1.4)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (2.9.0)
Requirement already satisfied: polars-runtime-32==1.35.2 in /usr/local/lib/python3.12/dist-packages (from polars>=0.20.0->ultralytics) (1.35.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (2025.1.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.29.3)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (4.12.0)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.4.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (2025.10.1)
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda12]>=12.8.0->ultralytics) (12.8.1)
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Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (9.19.0.56)
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Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (2.28.9)
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Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda12]>=12.8.0->ultralytics) (12.8.90)
Requirement already satisfied: nvidia-cuffrt-cu12==11.3.3.83.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda12]>=12.8.0->ultralytics) (11.3.3.83)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda12]>=12.8.0->ultralytics) (1.13.1.3)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda12]>=12.8.0->ultralytics) (12.8.90)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda12]>=12.8.0->ultralytics) (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda12]>=12.8.0->ultralytics) (11.7.3.90)
Requirement already satisfied: nvidia-cusparselt-cu12==12.5.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda12]>=12.8.0->ultralytics) (12.5.8.93)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cuda12]>=12.8.0->ultralytics) (12.8.93)
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Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings<13,>=12.9.4->ultralytics) (1.1.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.3.0->ultralytics) (1.17.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.8.0->ultralytics) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch>=1.8.0->ultralytics) (3.0.2)
Downloading ultralytics-8.4.75-py3-none-any.whl (1.3 MB)
```

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1.3/1.3 MB 71.0 MB/s eta 0:00:00
```

```
Downloading ultralytics_thop-2.0.20-py3-none-any.whl (28 kB)
```

```
Installing collected packages: ultralytics-thop, ultralytics
```

```
Successfully installed ultralytics-8.4.75 ultralytics-thop-2.0.20
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
!unzip "/content/drive/MyDrive/dataset.zip" -d /content/
```

```

inflating: /content/dataset/train/labels/image672.txt
inflating: /content/dataset/train/labels/image672.xml
inflating: /content/dataset/train/labels/image706.txt
inflating: /content/dataset/train/labels/image706.xml
inflating: /content/dataset/train/labels/image707.txt
inflating: /content/dataset/train/labels/image707.xml
inflating: /content/dataset/train/labels/image710.xml
inflating: /content/dataset/train/labels/image72.txt
inflating: /content/dataset/train/labels/image72.xml
inflating: /content/dataset/train/labels/image795.txt
inflating: /content/dataset/train/labels/image795.xml
inflating: /content/dataset/train/labels/image797.txt
inflating: /content/dataset/train/labels/image797.xml
inflating: /content/dataset/train/labels/image799.txt
inflating: /content/dataset/train/labels/image799.xml
inflating: /content/dataset/train/labels/image801.txt
inflating: /content/dataset/train/labels/image801.xml
inflating: /content/dataset/train/labels/image812.txt
inflating: /content/dataset/train/labels/image812.xml
inflating: /content/dataset/train/labels/image813.txt
inflating: /content/dataset/train/labels/image813.xml
inflating: /content/dataset/train/labels/image842.txt
inflating: /content/dataset/train/labels/image842.xml
inflating: /content/dataset/train/labels/image843.txt
inflating: /content/dataset/train/labels.cache
creating: /content/dataset/val/
creating: /content/dataset/val/images/
inflating: /content/dataset/val/images/image843.jpg
inflating: /content/dataset/val/images/image844.jpg
inflating: /content/dataset/val/images/image845.jpg
inflating: /content/dataset/val/images/image847.jpeg
inflating: /content/dataset/val/images/image950.jpeg
inflating: /content/dataset/val/images/image951.jpg
creating: /content/dataset/val/labels/
inflating: /content/dataset/val/labels/image843.xml
inflating: /content/dataset/val/labels/image844.txt
inflating: /content/dataset/val/labels/image844.xml
inflating: /content/dataset/val/labels/image845.txt
inflating: /content/dataset/val/labels/image845.xml
inflating: /content/dataset/val/labels/image847.txt
inflating: /content/dataset/val/labels/image847.xml
inflating: /content/dataset/val/labels/image950.txt
inflating: /content/dataset/val/labels/image950.xml
inflating: /content/dataset/val/labels/image951.txt
inflating: /content/dataset/val/labels/image951.xml
inflating: /content/dataset/val/labels.cache

```

```
yaml_path = "/content/dataset/data_custom.yaml"
```

```

with open(yaml_path, "w") as f:
    f.write("""
train: /content/dataset/train/images
val: /content/dataset/val/images

nc: 2
names: ["hat", "jacket"]
""")

```

```
!pip uninstall -y torch torchvision torchaudio
```

```

Found existing installation: torch 2.11.0+cu128
Uninstalling torch-2.11.0+cu128:
  Successfully uninstalled torch-2.11.0+cu128
Found existing installation: torchvision 0.26.0+cu128
Uninstalling torchvision-0.26.0+cu128:
  Successfully uninstalled torchvision-0.26.0+cu128
Found existing installation: torchaudio 2.11.0+cu128
Uninstalling torchaudio-2.11.0+cu128:
  Successfully uninstalled torchaudio-2.11.0+cu128

```

```
!pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu121
```

```
Uninstalling sympy-1.14.0:
  Successfully uninstalled sympy-1.14.0
Attempting uninstall: nvidia-nvtx-cu12
  Found existing installation: nvidia-nvtx-cu12 12.8.90
  Uninstalling nvidia-nvtx-cu12-12.8.90:
    Successfully uninstalled nvidia-nvtx-cu12-12.8.90
Attempting uninstall: nvidia-nccl-cu12
  Found existing installation: nvidia-nccl-cu12 2.28.9
  Uninstalling nvidia-nccl-cu12-2.28.9:
    Successfully uninstalled nvidia-nccl-cu12-2.28.9
Attempting uninstall: nvidia-cuspars-cu12
  Found existing installation: nvidia-cuspars-cu12 12.5.8.93
  Uninstalling nvidia-cuspars-cu12-12.5.8.93:
    Successfully uninstalled nvidia-cuspars-cu12-12.5.8.93
Attempting uninstall: nvidia-curand-cu12
  Found existing installation: nvidia-curand-cu12 10.3.9.90
  Uninstalling nvidia-curand-cu12-10.3.9.90:
    Successfully uninstalled nvidia-curand-cu12-10.3.9.90
Attempting uninstall: nvidia-cufft-cu12
  Found existing installation: nvidia-cufft-cu12 11.3.3.83
  Uninstalling nvidia-cufft-cu12-11.3.3.83:
    Successfully uninstalled nvidia-cufft-cu12-11.3.3.83
Attempting uninstall: nvidia-cuda-runtime-cu12
  Found existing installation: nvidia-cuda-runtime-cu12 12.8.90
  Uninstalling nvidia-cuda-runtime-cu12-12.8.90:
    Successfully uninstalled nvidia-cuda-runtime-cu12-12.8.90
Attempting uninstall: nvidia-cuda-nvrtc-cu12
  Found existing installation: nvidia-cuda-nvrtc-cu12 12.8.93
  Uninstalling nvidia-cuda-nvrtc-cu12-12.8.93:
    Successfully uninstalled nvidia-cuda-nvrtc-cu12-12.8.93
Attempting uninstall: nvidia-cuda-cupti-cu12
  Found existing installation: nvidia-cuda-cupti-cu12 12.8.90
  Uninstalling nvidia-cuda-cupti-cu12-12.8.90:
    Successfully uninstalled nvidia-cuda-cupti-cu12-12.8.90
Attempting uninstall: nvidia-cublas-cu12
  Found existing installation: nvidia-cublas-cu12 12.8.4.1
  Uninstalling nvidia-cublas-cu12-12.8.4.1:
    Successfully uninstalled nvidia-cublas-cu12-12.8.4.1
Attempting uninstall: nvidia-cusolver-cu12
  Found existing installation: nvidia-cusolver-cu12 11.7.3.90
  Uninstalling nvidia-cusolver-cu12-11.7.3.90:
    Successfully uninstalled nvidia-cusolver-cu12-11.7.3.90
Attempting uninstall: nvidia-cudnn-cu12
  Found existing installation: nvidia-cudnn-cu12 9.19.0.56
  Uninstalling nvidia-cudnn-cu12-9.19.0.56:
    Successfully uninstalled nvidia-cudnn-cu12-9.19.0.56
Successfully installed nvidia-cublas-cu12-12.1.1.105 nvidia-cuda-cupti-cu12-12.1.105 nvidia-cuda-nvrtc-cu12-12.1.105 nvidia-cuda-
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!pip install ultralytics
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Requirement already satisfied: ultralytics in /usr/local/lib/python3.12/dist-packages (8.4.75)
Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.0.2)
Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (3.10.0)
Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (4.13.0.92)
Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (11.3.0)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (6.0.3)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.32.4)
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.16.3)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.5.1+cu121)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (0.20.1+cu121)
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (5.9.5)
Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.35.2)
Requirement already satisfied: nvidia-ml-py>=12.0.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (13.610.43)
Requirement already satisfied: ultralytics-thop>=2.0.18 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.0.20)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.3.0)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (4.53.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (3.0.7)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (24.1)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (3.1.4)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (2.9.0)
Requirement already satisfied: polars-runtime-32==1.35.2 in /usr/local/lib/python3.12/dist-packages (from polars>=0.20.0->ultralytics) (1.35.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (2025.1.1)
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Requirement already satisfied: typing-extensions>=4.8.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (4.12.2)
Requirement already satisfied: networkx in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.4.2)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.1.4)
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Requirement already satisfied: nvidia-cudnn-cu12==9.1.0.70 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral
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Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral
Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral
Requirement already satisfied: nvidia-nvjitlink-cu12 in /usr/local/lib/python3.12/dist-packages (from nvidia-cusolver-cu12==11.4
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy==1.13.1->torch>=1.8.0->
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.3.0
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.8.0->ultral
```

```
import torch
print(torch.__version__)
print(torch.cuda.is_available())
```

```
2.5.1+cu121
True
```

```
from ultralytics import YOLO

# load model
model = YOLO("yolov8m.pt")

# train
model.train(
    data="/content/dataset/data_custom.yaml",
    epochs=100,
    imgsz=640,
    batch=8,
    task="detect"
)
```

```
[
    1,
    1,
    0.875, ...,
    0,
    0,
    0]], 'Confidence', 'Recall']]
fitness: 0.40675234140859134
keys: ['metrics/precision(B)', 'metrics/recall(B)', 'metrics/mAP50(B)', 'metrics/mAP50-95(B)']
maps: array([
    0.42759,
    0.38591])
names: {0: 'hat', 1: 'jacket'}
nt_per_class: array([10,  8])
nt_per_image: array([5,  4])
results_dict: {'metrics/precision(B)': 0.7462687700298343, 'metrics/recall(B)': 0.875, 'metrics/mAP50(B)': 0.8642694805194804,
'metrics/mAP50-95(B)': 0.40675234140859134, 'fitness': 0.40675234140859134}
save_dir: PosixPath('/content/runs/detect/train')
speed: {'preprocess': 0.22667516668661847, 'inference': 15.093155999996574, 'loss': 0.0020806666422383082, 'postprocess':
1.9246703333237747}
stats: {'tp': [], 'conf': [], 'pred_cls': [], 'target_cls': [], 'target_img': []}
```

```
!yolo task=detect mode=predict \
model=/content/runs/detect/train/weights/yolov8m_custom.pt \
show=True \
conf=0.5 \
source=/content/1.jpg
```

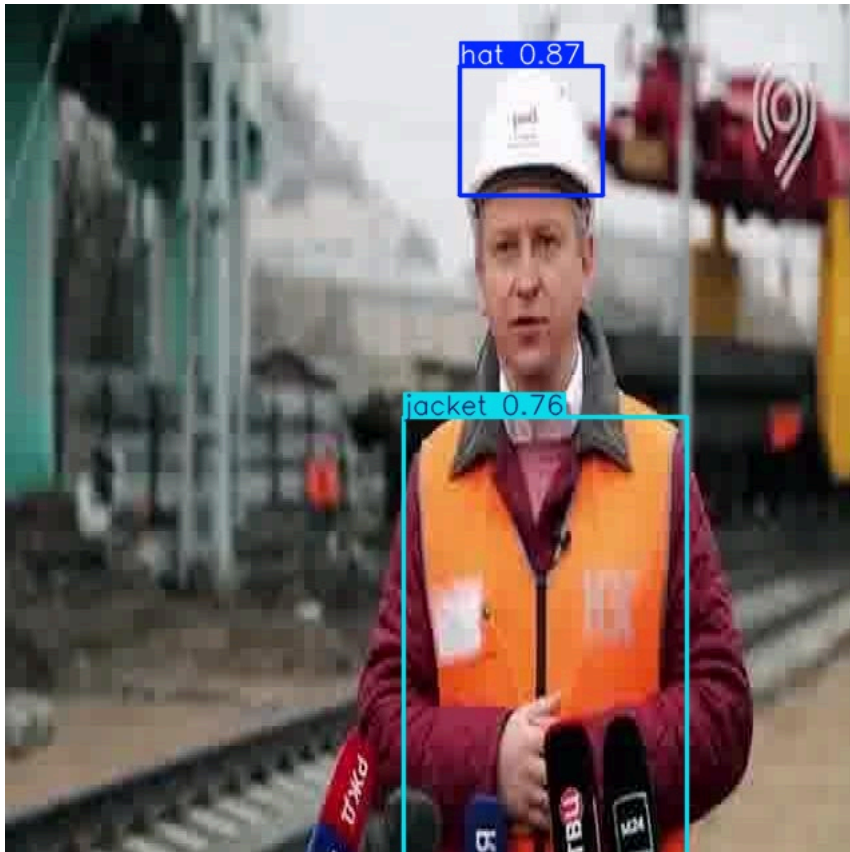
WARNING ⚠ Environment does not support cv2.imshow() or PIL Image.show()

Ultralytics 8.4.75 🚀 Python-3.12.13 torch-2.5.1+cu121 CUDA:0 (Tesla T4, 14913MiB)
Model summary (fused): 93 layers, 25,840,918 parameters, 0 gradients, 78.7 GFLOPs

image 1/1 /content/1.jpg: 640x640 1 hat, 1 jacket, 35.5ms
Speed: 2.9ms preprocess, 35.5ms inference, 16.3ms postprocess per image at shape (1, 3, 640, 640)
Results saved to /content/runs/detect/predict
🔗 Learn more at <https://docs.ultralytics.com/modes/predict>

```
from IPython.display import Image, display

display(Image("/content/runs/detect/predict/1.jpg"))
```



```
!yolo task=detect mode=predict \
model=/content/runs/detect/train/weights/yolov8m_custom.pt \
source=/content/construction_workers.mp4 \
conf=0.5 \
save=True
```

video 1/1 (from 257/100) /content/construction_workers.mp4: 284x640 (pre-detections) 13.0ms

```

video 1/1 (frame 357/409) /content/construction_workers.mp4: 384x640 (no detections), 13.0ms
video 1/1 (frame 358/409) /content/construction_workers.mp4: 384x640 (no detections), 13.4ms
video 1/1 (frame 359/409) /content/construction_workers.mp4: 384x640 (no detections), 13.6ms
video 1/1 (frame 360/409) /content/construction_workers.mp4: 384x640 (no detections), 14.2ms
video 1/1 (frame 361/409) /content/construction_workers.mp4: 384x640 (no detections), 15.1ms
video 1/1 (frame 362/409) /content/construction_workers.mp4: 384x640 (no detections), 15.4ms
video 1/1 (frame 363/409) /content/construction_workers.mp4: 384x640 (no detections), 15.0ms
video 1/1 (frame 364/409) /content/construction_workers.mp4: 384x640 (no detections), 14.5ms
video 1/1 (frame 365/409) /content/construction_workers.mp4: 384x640 (no detections), 13.8ms
video 1/1 (frame 366/409) /content/construction_workers.mp4: 384x640 (no detections), 13.3ms
video 1/1 (frame 367/409) /content/construction_workers.mp4: 384x640 (no detections), 12.7ms
video 1/1 (frame 368/409) /content/construction_workers.mp4: 384x640 (no detections), 12.4ms
video 1/1 (frame 369/409) /content/construction_workers.mp4: 384x640 (no detections), 12.5ms
video 1/1 (frame 370/409) /content/construction_workers.mp4: 384x640 (no detections), 12.4ms
video 1/1 (frame 371/409) /content/construction_workers.mp4: 384x640 (no detections), 12.6ms
video 1/1 (frame 372/409) /content/construction_workers.mp4: 384x640 (no detections), 12.7ms
video 1/1 (frame 373/409) /content/construction_workers.mp4: 384x640 (no detections), 12.8ms
video 1/1 (frame 374/409) /content/construction_workers.mp4: 384x640 (no detections), 13.1ms
video 1/1 (frame 375/409) /content/construction_workers.mp4: 384x640 (no detections), 13.2ms
video 1/1 (frame 376/409) /content/construction_workers.mp4: 384x640 (no detections), 13.3ms
video 1/1 (frame 377/409) /content/construction_workers.mp4: 384x640 (no detections), 13.7ms
video 1/1 (frame 378/409) /content/construction_workers.mp4: 384x640 (no detections), 14.1ms
video 1/1 (frame 379/409) /content/construction_workers.mp4: 384x640 (no detections), 14.7ms
video 1/1 (frame 380/409) /content/construction_workers.mp4: 384x640 (no detections), 15.1ms
video 1/1 (frame 381/409) /content/construction_workers.mp4: 384x640 (no detections), 15.1ms
video 1/1 (frame 382/409) /content/construction_workers.mp4: 384x640 (no detections), 14.8ms
video 1/1 (frame 383/409) /content/construction_workers.mp4: 384x640 (no detections), 14.0ms
video 1/1 (frame 384/409) /content/construction_workers.mp4: 384x640 (no detections), 13.2ms
video 1/1 (frame 385/409) /content/construction_workers.mp4: 384x640 (no detections), 12.7ms
video 1/1 (frame 386/409) /content/construction_workers.mp4: 384x640 (no detections), 12.5ms
video 1/1 (frame 387/409) /content/construction_workers.mp4: 384x640 (no detections), 12.6ms
video 1/1 (frame 388/409) /content/construction_workers.mp4: 384x640 (no detections), 12.7ms
video 1/1 (frame 389/409) /content/construction_workers.mp4: 384x640 (no detections), 12.7ms
video 1/1 (frame 390/409) /content/construction_workers.mp4: 384x640 (no detections), 13.0ms
video 1/1 (frame 391/409) /content/construction_workers.mp4: 384x640 (no detections), 13.5ms
video 1/1 (frame 392/409) /content/construction_workers.mp4: 384x640 (no detections), 14.3ms
video 1/1 (frame 393/409) /content/construction_workers.mp4: 384x640 (no detections), 16.3ms
video 1/1 (frame 394/409) /content/construction_workers.mp4: 384x640 (no detections), 14.8ms
video 1/1 (frame 395/409) /content/construction_workers.mp4: 384x640 (no detections), 13.0ms
video 1/1 (frame 396/409) /content/construction_workers.mp4: 384x640 (no detections), 12.6ms
video 1/1 (frame 397/409) /content/construction_workers.mp4: 384x640 (no detections), 12.5ms
video 1/1 (frame 398/409) /content/construction_workers.mp4: 384x640 (no detections), 12.5ms
video 1/1 (frame 399/409) /content/construction_workers.mp4: 384x640 (no detections), 12.8ms
video 1/1 (frame 400/409) /content/construction_workers.mp4: 384x640 (no detections), 13.3ms
video 1/1 (frame 401/409) /content/construction_workers.mp4: 384x640 (no detections), 13.6ms
video 1/1 (frame 402/409) /content/construction_workers.mp4: 384x640 (no detections), 14.2ms
video 1/1 (frame 403/409) /content/construction_workers.mp4: 384x640 (no detections), 15.3ms
video 1/1 (frame 404/409) /content/construction_workers.mp4: 384x640 (no detections), 15.7ms
video 1/1 (frame 405/409) /content/construction_workers.mp4: 384x640 (no detections), 15.3ms
video 1/1 (frame 406/409) /content/construction_workers.mp4: 384x640 (no detections), 14.7ms
video 1/1 (frame 407/409) /content/construction_workers.mp4: 384x640 (no detections), 14.0ms
video 1/1 (frame 408/409) /content/construction_workers.mp4: 384x640 (no detections), 13.7ms
video 1/1 (frame 409/409) /content/construction_workers.mp4: 384x640 (no detections), 13.2ms
Speed: 2.5ms preprocess, 13.7ms inference, 1.1ms postprocess per image at shape (1, 3, 384, 640)
Results saved to /content/runs/detect/predict-2

```

[Learn more at https://docs.ultralytics.com/modes/predict](https://docs.ultralytics.com/modes/predict)

```

import os

print(os.listdir("/content/runs/detect/predict-2"))

['construction_workers.avi']

```

```

from IPython.display import HTML
from base64 import b64encode

video_path = "/content/runs/detect/predict-2/construction_workers.avi"

mp4 = open(video_path, "rb").read()
data_url = "data:video/avi;base64," + b64encode(mp4).decode()

HTML(f"""
<video width=700 controls>
  <source src="{data_url}" type="video/avi">
</video>
""")

```